9.1 Composition of carbohydrates

Carbohydrates are so called because they contain carbon with hydrogen and oxygen in the same proportion as in water (2:1). The general formula is $C_nH_{2n}O_n$

9.2 Classification of carbohydrates

Carbohydrates are classified according to the number of saccharide (sugar) groups present. They are broadly classified as simple carbohydrates and complex carbohydrates. The simple carbohydrates include monosaccharides(Single sugar) and disaccharides (Double sugars). Complex carbohydrates include starch, glycogen and fibers. The classification of carbohydrates is schematically represented below:

9.2.1 Simple Carbohydrates

A) Monosaccharides

They have one saccharide group and are the simplest form of carbohydrates. All carbohydrates are reduced to this state before absorption and utilization. They contain 3-6 carbon atoms and are accordingly termed triose, tetrose, pentose or hexose.

(i) **Biose:** $C_2H_4O_2$ (e.g.)Glycolic aldehyde

(ii) **Triose:** $C_3H_6O_3$ (e.g.)Glyceraldehyde and Dihydroxyacetone. They occur in plant and animal tissues in small amounts and are derived from the breakdown of glucose.

(iii) **Tetroses:** $C_4H_8O_4$ (e.g.) Erythrose, Threose

(iv) **Pentoses:** $C_5H_{10}O_5$(e.g.) Arabinose, Xylose, Ribose and Deoxyribose.

(v) **Hexoses:** $C_6H_{12}O_6$ .They are further sub-divided into 2 groups(i.e) Aldoses or sugars containing aldehyde group(e.g.) Glucose,

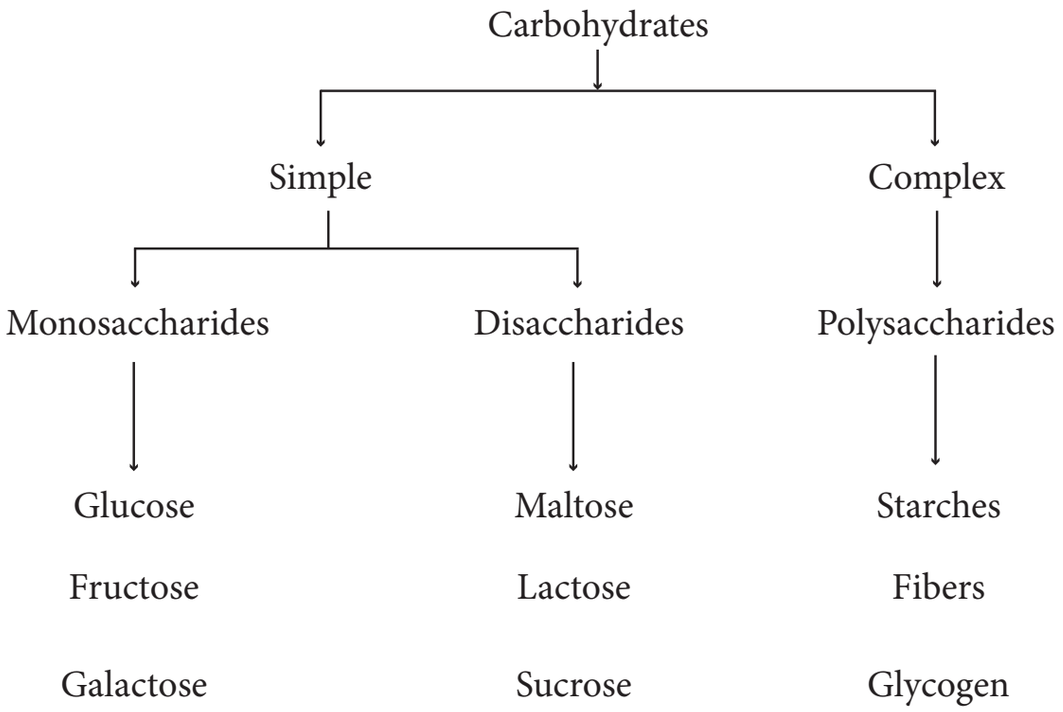


Fig 9.1: Classification of carbohydrates

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